ISA Transactions 56 (2015) 276–287

Contents lists available at ScienceDirect

ISA Transactions

journal homepage: www.elsevier.com/locate/isatrans

Design and implementation of adaptive PI control schemes for web tension control in roll-to-roll (R2R) manufacturing

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ARTICLE INFO

Article history:
Received 4 April 2014
Received in revised form
28 October 2014
Accepted 30 November 2014
Available online 30 December 2014
This paper was recommended for publication by Dr. Didier Theilliol

Keywords:
Adaptive control
Roll-to-roll manufacturing
PI control
Web tension
Relay feedback technique
Web handling systems

ABSTRACT

In this paper, two adaptive Proportional-Integral (PI) control schemes are designed and discussed for control of web tension in Roll-to-Roll (R2R) manufacturing systems. R2R systems are used to transport continuous materials (called webs) on rollers from the unwind roll to the rewind roll. Maintaining web tension at the desired value is critical to many R2R processes such as printing, coating, lamination, etc. Existing fixed gain PI tension control schemes currently used in industrial practice require extensive tuning and do not provide the desired performance for changing operating conditions and material properties. The first adaptive PI scheme utilizes the model reference approach where the controller gains are estimated based on matching of the actual closed-loop tension control systems with an appropriately chosen reference model. The second adaptive PI scheme utilizes the indirect adaptive control approach together with relay feedback technique to automatically initialize the adaptive PI gains. These adaptive tension control schemes can be implemented on any R2R manufacturing system. The key features of the two adaptive schemes is that their designs are simple for practicing engineers, easy to implement in real-time, and automate the tuning process. Extensive experiments are conducted on a large experimental R2R machine which mimics many features of an industrial R2R machine. These experiments include trials with two different polymer webs and a variety of operating conditions. Implementation guidelines are provided for both adaptive schemes. Experimental results comparing the two adaptive schemes and a fixed gain PI tension control scheme used in industrial practice are provided and discussed.

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1. Introduction

In Roll-to-Roll (R2R) manufacturing of continuous materials, control of web tension is critical during transport of the materials through many processes such as printing, coating, and lamination. It plays an important role in ensuring the quality of the finished web products. During transport, many processes may require transport under different operating speeds and heating/cooling of the webs. Further, there are many process and machine induced disturbances which the tension control systems must compensate to achieve a good tension regulation performance. Tension variations cause a variety of product defects. For example, high web tension may result in wrinkles and tears while low tension may cause web slack, loss of traction on rollers during transport, and difficulty in guiding the webs on rollers. The consolidated effect of large web tension variations may result in disruption in production, diminishing product quality, and machine hardware damage.

Industrial web tension control systems typically employ a fixed gain PI controller. The PI controller gains are tuned to give a stable response for a given operating condition and material. This is typically done empirically on the machine because analytical tuning of PI controllers is a challenging task due to changing operating conditions and uncertainty in the knowledge of web material parameters and machine parameters. In practice, the PI tension controllers are tuned on-site based on real-time observation of the tension response performance. When operating conditions or material properties change, the fixed gain PI tension controllers do not provide adequate performance or in some cases render the closed-loop tension control system unstable. The better approach is to use an adaptive control scheme that facilitates the adjustment of controller parameters for such changing conditions. An adaptive controller can modify closed-loop system behavior by compensating for the changes in system parameters.

There are many approaches to design adaptive control schemes, such as gain scheduling, self-tuning regulators, and model reference adaptive control [1,2,8,13]. The direct and indirect adaptive schemes are promising but the design and implementation of those controllers are cumbersome and provide many difficulties for practicing engineers. For example, since many of these schemes are model based and

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http://dx.doi.org/10.1016/j.isatra.2014.11.020
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the web system dynamic model is involved, the designs are complicated and result in a large number of estimated parameters. Simple adaptive control structures which mimic most of the features of the fixed gain PI schemes are desirable. These schemes can provide the automation to circumvent tuning under a wide variety of system parameter changes, and yet provide the ease of design and implementation.

A R2R manufacturing machine is a large-scale system which can be simplified for analysis and control design purposes by decomposing it into a number of interconnected subsystems. There has been substantial reported work in the literature in modeling and control of R2R systems. The tension dynamic model that considers entering web span tension was first developed in King [10]. Brandenburg [5] and Shelton [19] assumed strain in the web to be very small and derived the governing equation for web tension in a span with the small strain assumption. An early overview of longitudinal and lateral dynamic behavior and tension control can be found in [21]. A self-tuning regulator for web tension regulation is given in [4]. A robust H_∞ controller is designed and implemented in [6,11]. Linearization of the web system dynamic model and development of a decentralized state feedback control scheme are given in [17]. A state space reference model based adaptive control technique is given in [15]; a special reference model is chosen based on the overall dynamics of the large-scale R2R system. Model reference direct and indirect adaptive schemes for web tension control are investigated and implemented on an experimental platform in [18]; the motivation for the design and development of simple adaptive PI schemes came from this work. Application of fault tolerant control to a winding machine is given in [14]. Tension control in the web is also critical in the presence of key primitive elements used in R2R manufacturing such as accumulators [16,12] and print cylinders.

In this paper, first a simple direct model reference adaptive PI (MRA-PI) controller based on the gradient method is designed. Controller gains are estimated by matching the plant performance and desired characteristics provided by a reference model. The estimates of the controller parameters are initialized by considering the stability of the nominal closed-loop tension control system. Second, an indirect adaptive PI control scheme that would facilitate automatic initialization of estimated parameters based on the relay feedback technique is investigated. The indirect adaptive PI control algorithm is simple and initialization of controller parameter estimates does not need the knowledge of nominal plant parameters. There are several ways to initialize the estimated parameters. The application of the relay feedback technique in industrial controllers is given in [7]. This technique is alternative to the conventional continuous cycling method used to generate sustained oscillations. The relay feedback method is known to be effective in determining the ultimate gain and ultimate frequency (frequency at which the phase shift of the plant is 180 degrees) of a system. An algorithm to compute the estimate of ultimate frequency in real-time is considered to account for its change due to plant parameter changes; the method given in [3] is employed where a small sinusoidal oscillation is introduced into the closed-loop tension control system. The fixed gain PI controller gains are tuned by comparing the closed-loop speed transfer function characteristic equation with a standard second-order characteristic equation with design parameters as the damping ratio and natural frequency. These gains are further tuned in experimentation to obtain the best possible performance. For complex processes, gain tuning methods of PID controller can be found in [20,9]. The two adaptive PI schemes together with an industrial fixed gain PI tension control scheme are implemented on a large experimental R2R platform containing multiple driven rollers and tension zones. The implementation guidelines together with the discussion of experimental results are provided.

The rest of the paper is organized as follows. Governing equations for web tension and transport speed are discussed in Section 2. The existing industrial tension control schemes are described in Section 3. The design of the MRA-PI controller is presented in Section 4. In Section 5, design of indirect adaptive PI based on the relay feedback technique is presented; estimation of plant transfer function around the ultimate frequency is also discussed. The experimental setup and guidelines to implement the MRA-PI and adaptive PI controller to any general web line are provided in Section 6. The performance of the adaptive PI control schemes is presented through experimental results in Section 7. Concluding remarks and future work are given in Section 8.

2. Governing equations for web tension and transport speed

A roll-to-roll system consists of key primitive elements such as material rolls, driven and idle rolls, and web span (web between two adjacent rollers). A number of these primitive elements are employed sequentially to construct an R2R system. The governing equation for these primitive elements is derived and composed as per the configuration of the R2R system to develop a mathematical model that can describe the transport behavior of webs. For example, a line sketch of an experimental R2R platform is shown in Fig. 1; the system consists of a number of driven and idle rollers and web spans between unwind and rewind rolls.

The governing equation for web speed on a roller is obtained by considering the angular motion of the roller and web wrap on that roller. And the governing equation for web tension in a span is obtained by first applying conservation of mass to a control volume encompassing the web between two adjacent rollers to obtain a governing equation for web strain. Then, a constitutive law is used to relate web strain and web tension. In this paper, the material is assumed to be linearly elastic and a linear relation is used via the modulus of elasticity of the material. The web material is transported at low strain (10%) and the web strain for which the material is elastic and satisfies the small strain assumption is provided in the shaded region of Fig. 2. A simple two roller setup with adjacent web spans is provided in Fig. 3 to illustrate a portion of any R2R system and show the nomenclature for presenting the governing equations below. It is assumed that web is not slipping on the roller during transport, that is, the peripheral velocity of the roller is assumed to be equal to the web velocity on the roller.

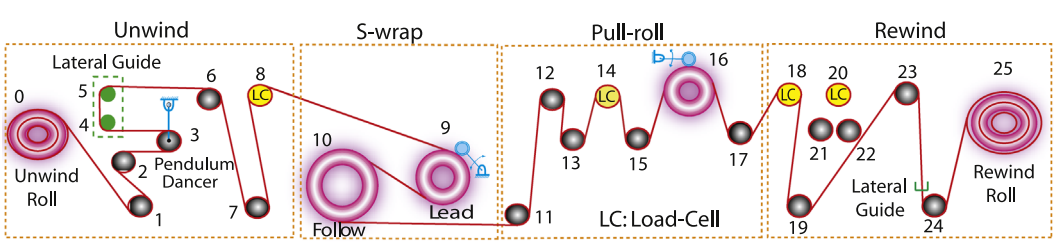

Fig. 1. Line sketch of R2R experimental platform.

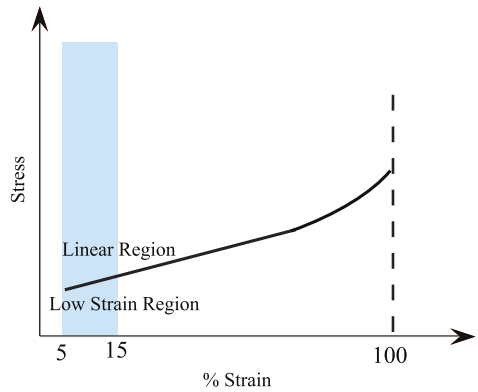

Fig. 2. Two roller setup.

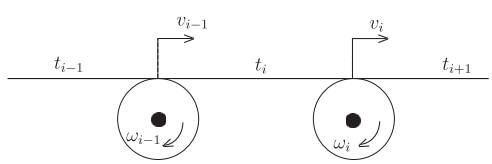

Fig. 3. Two roller setup.

In this setup, t_i denotes web tension in the span between $(i-1)$th and ith rollers, v_i is the web transport speed on the ith roller, and ω_i is the angular velocity of the ith roller. There has been significant reported work in the literature on modeling of web transport behavior and we will simply list the governing equations here and refer interested readers to the literature for more details.

The governing equation for web tension in the ith span is given by

$$\dot{t}_i = \frac{EA}{L_i}(v_i - v_{i-1}) + \frac{1}{L_i}(t_{i-1}v_{i-1} - t_i v_i), \tag{1}$$

where E is the modulus of elasticity of the web material, A is the area of cross-section of the web material, and L_i is the span length. The governing equation for web transport speed on the ith roller is given by

$$\dot{v}_i = \frac{R_i^2}{J_i}(t_{i+1} - t_i) + \frac{R_i}{J_i}n_i u_i - \frac{b_{fi}}{J_i}v_i. \tag{2}$$

where R_i is the radius of the ith roller, J_i is the inertia of the ith roller, n_i is the gear ratio, and b_{fi} is the coefficient of friction.

The above governing equations can be linearized around reference values. To do this, define the perturbations from references as $\Delta t_i = t_i - t_r$, $\Delta v_i = v_i - v_r$, and $\Delta u_i = u_i - u_{eq}$, where u_{eq} is the equilibrium control input. Therefore, the linearized governing equations in the frequency domain are given by

$$T_i(s) = \frac{EA/V_r}{\tau_{\omega i}s+1}(V_i(s) - V_{i-1}(s)) + \frac{1}{\tau_{\omega i-1}s+1}T_{i-1}(s), \tag{3}$$

$$V_i(s) = \frac{R_i^2}{J_i s + b_{fi}}(T_{i+1}(s) - T_i(s)) + \frac{R_i}{J_i s + b_{fi}}n_i U_i(s) \tag{4}$$

where $T_i(s)$ is the Laplace transform of $\Delta t_i(t)$, that is, $T_i(s) = \mathcal{L}\{\Delta t_i\}$, $V_i(s) = \mathcal{L}\{\Delta v_i\}$, $U_i(s) = \mathcal{L}\{\Delta u_i\}$, and $\tau_{\omega i} = L_i/v_r$ is the span time constant, that is, the time it takes for an element of material from entry to exit of that span. The term containing $T_{i-1}(s)$ in the tension governing equation appears due to transport of strain from upstream spans to downstream spans. From these two governing equations one can create a model for the entire R2R transport system by writing down web speed on each roller and web tension in each span. Many idle rollers (non-driven) are employed in addition to driven rollers to create web paths through processes

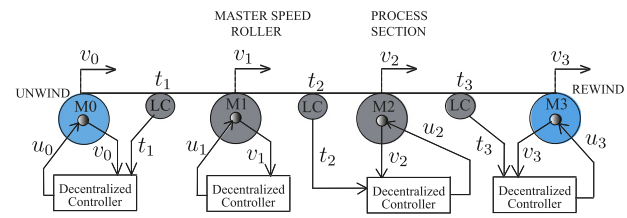

Fig. 4. Simplified line sketch of R2R experimental platform.

and to provide support for the web during transport. In the presence of a large number of idle rollers the transport behavior model for the system is too cumbersome. Often the notion of tension zone is employed in practice to simplify the models as the idle rollers do not contribute much to the dynamics during steady state operation. The idea is to designate the web between any two driven rollers as a tension zone and consider each zone to be a web span. This simplifies the dynamic model significantly. A simplified line sketch showing the tension zones and driven rollers is provided in Fig. 4; the rollers indicated as "LC" are mounted on load-cells to provide tension feedback for tension control systems. The two driven rollers in the S-wrap section are electronically slaved together and are under pure speed control; these are treated as a one driven roller (M1) in the simplified sketch and are typically referred to as the Master Speed Roller.

3. Industrial tension control schemes

Two strategies are mainly used for web tension control; load-cell based and dancer based feedback control systems. In a load-cell based tension control scheme, web tension measured by load cells mounted on idle rollers is used as feedback for the control system. In the dancer based scheme, displacement of the dancer (either linear or rotational) due to web tension variations is sensed by a rotary variable differential transducer and used as feedback for the control system.

As indicated in Fig. 4 web tension in each tension zone is controlled by the driven roller adjacent to the zone, i.e., a decentralized controller which uses sensed information from that particular zone is typically employed in almost all industrial web tension control systems. This provides simplicity in implementation as well as the ability to isolate and diagnose problems in each zone in an efficient manner. Two types of web tension control schemes are employed in industrial practice. (1) Torque Based Tension Control System: The output of the tension controller provides a correction to the motor torque reference; a sketch is provided in Fig. 5 with switch at B. (2) Speed Based Tension Control System: The output of the tension controller provides correction to the speed reference; a sketch is provided in Fig. 5 with switch at A. Most modern web tension control systems employ the speed-based tension control scheme because of the motor drive's ability to provide precise speed control. Therefore, the work in this paper will focus on speed based tension control systems.

In web tension control systems, web tension is typically measured by an idle roller mounted on load cells within the tension zone or by inferring tension from the displacement of a device called the dancer system. The dancer system consists of a pneumatically or hydraulically loaded oscillating roller (either linear or rotational) whose oscillatory movement is generated due to variations in web tension. In this paper, load cell rollers are used to measure web tension. A comparative discussion of load-cell based versus dancer based tension control systems is given in [18].

In the following we discuss the development of two adaptive PI tension controllers which can be used for any tension zone.

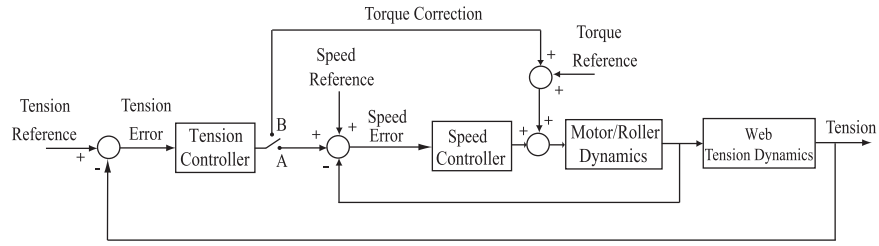

Fig. 5. Torque and speed based tension control schemes.

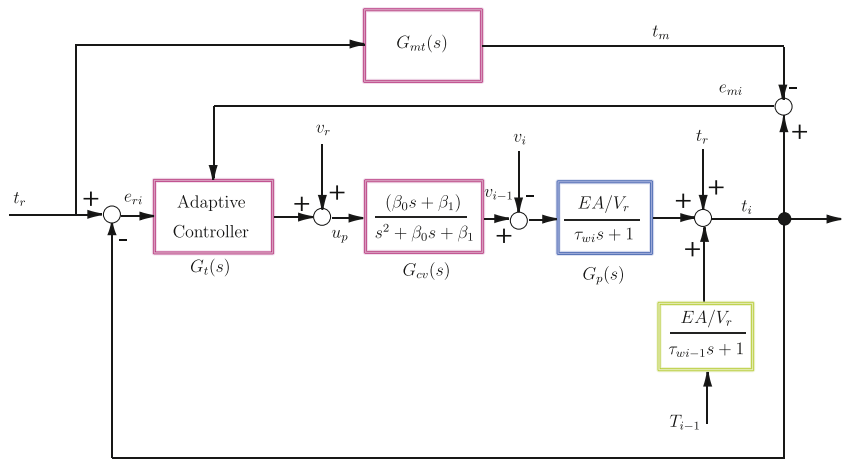

Fig. 6. Model reference adaptive PI tension control strategy.

4. Control scheme I: Model reference adaptive proportional integral (MRA-PI) controller

In the MRA-PI control, the plant is parameterized in terms of the PI controller parameters and controller parameters are estimated online using parameter adaptation laws. The goal is to find a parameter adjustment mechanism to achieve zero error between the reference model output and the actual system output; tension is the output for our application. For the speed loop we use an industrial PI scheme that is implemented on the motor drive. The open loop transfer function for the speed loop is given by

$$G_v(s)=\frac{k_v(s+\omega_v)}{s}\frac{1}{n_{i-1}J_{i-1}s} \tag{5}$$

where k_v is the proportional gain and ω_v is the cutoff frequency. The closed-loop transfer function for the inner velocity loop takes the form of

$$G_{cv}(s)=\frac{(\beta_0 s+\beta_1)}{s^2+\beta_0 s+\beta_1} \tag{6}$$

where $\beta_0=k_v/(n_{i-1}J_{i-1})$, $\beta_1=\beta_0\omega_v$, $\omega_v=\beta_0/(4\zeta^2)$, ζ is the damping ratio of the system.

The structure of the MRA-PI tension control scheme is shown in Fig. 6. The outer loop fixed gain PI web tension controller is replaced by the MRA-PI controller as shown in Fig. 6. The open loop transfer function from the velocity correction (output of the tension controller) to the tension variation (output of $G_p(s)$) is $G_{cv}G_p$. The closed-loop transfer function with the MRI-PI controller is given by

$$G_{ct}(s)=\frac{G_tG_{cv}G_p}{1+G_tG_{cv}G_p}. \tag{7}$$

where G_t is the adaptive PI transfer function.

The selection of the structure of the reference model is important. We choose a structure for the reference model that is similar to the closed-loop system obtained by using two fixed gain PI controllers for the speed and tension loops. The relative degree of this closed loop transfer function is two. We select a reference model with relative degree two and having the same number of poles and zeros as the closed-loop transfer function resulting from the fixed gain PI controllers. This is given by

$$G_{mt}(s)=\frac{k_m(as+z_1)(s+z_2)}{(s+p_1)(s+p_2)(s+p_3)(s+p_4)} \tag{8}$$

where the reference model parameters k_m, a, z_i and p_i are selected to provide the desired performance through model simulations of the MRA-PI tension control scheme. As it is typically done in model reference adaptive schemes, we develop parameter adaptation laws for the tension controller PI gains by minimizing the loss function corresponding to the model error $e_{mi}=t_i-t_m$. The MRA-PI controller has two adjustable parameters, proportional and integral gains. Define the parameter vector as $\theta=[k_p\ k_i]$. The parameter adaptation law by using the gradient method is given by

$$\frac{d\theta}{dt}=-\gamma e_{mi}\frac{\partial e_{mi}}{\partial\theta} \tag{9}$$

where $\partial e_{mi}/\partial\theta$ is the sensitivity of the model error to parameter θ. Since the reference model output does not depend on the controller parameters, the sensitivity is given by

$$\frac{\partial e_{mi}}{\partial\theta}=\frac{\partial(t_i-t_m)}{\partial\theta}=\frac{\partial t_i}{\partial\theta}. \tag{10}$$

Therefore, the resulting adaptation laws are

$$\frac{dk_p}{dt}=-\gamma_p e_{mi}\frac{s(EA/V_r)(\beta_0 s+\beta_1)}{s(s^2+\beta_0 s+\beta_1)(\tau_{\omega i}s+1)+(k_p s+k_i)(EA/V_r)(\beta_0 s+\beta_1)}, \tag{11}$$

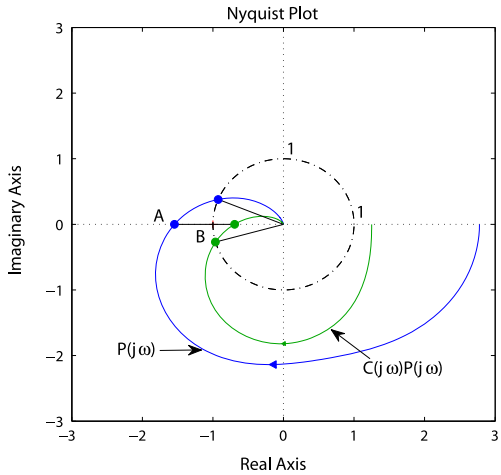

Fig. 7. Example Nyquist plot for a plant with and without compensator.

$$\frac{dk_i}{dt} = -\gamma_i e_{mi} \frac{(EA/V_r)(\beta_0 s+\beta_1)}{s(s^2+\beta_0 s+\beta_1)(\tau_{\omega i} s+1)+(k_p s+k_i)(EA/V_r)(\beta_0 s+\beta_1)} \tag{12}$$

where γ_p and γ_i are adaptive gains. Since the system parameters are not known, the above adaptation laws cannot be used. The following adaptation laws that utilize the reference model structure and parameters is used instead:

$$\frac{dk_p}{dt} = -\gamma_p e_{mi} \frac{k_m s(s+z_2)}{(s+p_1)(s+p_2)(s+p_3)(s+p_4)}, \tag{13}$$

$$\frac{dk_i}{dt} = -\gamma_i e_{mi} \frac{k_m (s+z_2)}{(s+p_1)(s+p_2)(s+p_3)(s+p_4)}. \tag{14}$$

For online adaptation in real-time control, one requires good initialization for the controller parameter estimates. We consider the closed-loop characteristic equation with MRA-PI which is given by

$$s(s^2+\beta_0 s+\beta_1)(\tau_{\omega i} s+1)+(EA/V_r)(\beta_0 s+\beta_1)(k_p s+k_i) = 0. \tag{15}$$

To obtain constraints on the gains k_p and k_i which provide an idea on how to initialize the estimated parameters, we apply Hurwitz criteria for the roots of the characteristic equation to be in the open left half of the complex plane, which gives the following constraints:

$$k_p > \frac{(1+\omega_v \tau_{\omega i})}{(EA/V_r)}, \tag{16}$$

$$k_i > -\frac{\omega_v}{(EA/V_r)} - k_p \omega_v. \tag{17}$$

Note that the right-hand side of these constraints themselves require the system parameters. We use nominal system parameters to initialize the estimated PI controller parameters.

5. Control scheme II: Indirect adaptive PI control based on the relay feedback technique

An indirect adaptive PI control scheme is described in this section which does not require the knowledge of the nominal parameters of the model. The relay feedback technique is first employed to find the amplitude of the plant and the frequency at which the phase of the plant is 180 degrees. The parameters of the plant transfer function are estimated first and a controller is selected to provide the required gain and phase margins. The advantage of this indirect adaptive PI controller is that one can automatically initialize the controller parameters. This is particularly useful for R2R manufacturing systems where it is difficult to know the nominal values of the plant parameters. We describe the method in the following.

Let $P(s)$ be the transfer function of the plant to be controlled and $C(s)$ be the adaptive PI used to control the plant. The Nyquist plots of the plant and the open loop system along with the controller and the plant are shown in Fig. 7.

In traditional frequency response based controller design, if we have knowledge about the plant, one can obtain its frequency response and the controller can be designed by shaping the frequency response to the desired. If the frequency response of the plant is not known, one cannot use this approach. In such a scenario, if the magnitude and frequency corresponding to point A of the frequency response of $P(s)$ are available, then the plant transfer function can be estimated (from the information of point A) which can be used to design the controller. Let $C(s)$ be the controller transfer function and B be the point on the Nyquist plot of the open loop system ($C(s)P(s)$) with the desired specifications (gain margin and phase margin) as indicated in Fig. 7. The goal is to design the controller $C(s)$ such that point A corresponding to the plant transfer function can be moved to point B corresponding to the open loop transfer function. In order to determine point B, the magnitude of the plant transfer function and associated frequency at point A (ω_A) need to be identified. This frequency ω_A is typically referred to in the literature as the ultimate frequency.

The Relay feedback technique may be employed to obtain the ultimate frequency and the magnitude of the plant transfer function at point A. The relay feedback technique can be readily applied to an R2R system. Introduction of the relay into the outer tension loop is shown in Fig. 8. A relay with hysteresis is employed to avoid high frequency switching of the standard relay. The relay logic is given by

If $e \geq \delta$, then $d = +5\%$ of reference variable;
If $e \leq -\delta$, then $d = -5\%$ of reference variable;
Else keep the previous output of relay.

where $e = t_i - t_r$ is the tension error, δ is the relay hysteresis, and d is the relay amplitude. The relay feedback method induces an oscillation in the controlled tension output which is used to identify the ultimate frequency and the plant amplitude at that frequency.

The indirect adaptive PI control scheme is illustrated in Fig. 9. The plant $G_p(s)$ is a combined representation of the tension dynamics and inner velocity closed-loop transfer function whose frequency response can be represented as

$$G_p(j\omega) = \alpha e^{j\phi} \tag{18}$$

where α is the amplitude and ϕ is the phase shift of the plant. Since the relative degree of $G_p(s)$ is two, this can be approximated by a second-order discrete model as

$$\bar{t}_i(t) = b_1 \bar{u}(t-T) + b_2 \bar{u}(t-2T) \tag{19}$$

where $\bar{u}$ and $\bar{t}_i$ are the filtered control input and process tension output, respectively; b_1 and b_2 are the unknown parameters to be estimated; and T is the sampling time.

The parameters of $G_p(j\omega)$ are estimated in real-time by using the least squares method. In order to estimate the plant, the control input and plant output are filtered using the band-pass filter

$$G_f(s) = \frac{s}{s^2+2\zeta\omega_f s+\omega_f^2}, \tag{20}$$

where ω_f is the band pass filter frequency and can be chosen as the estimated ultimate frequency. The ratio of the filtered output amplitude to the filtered input amplitude is the plant magnitude α. The phase difference between the filtered output and input

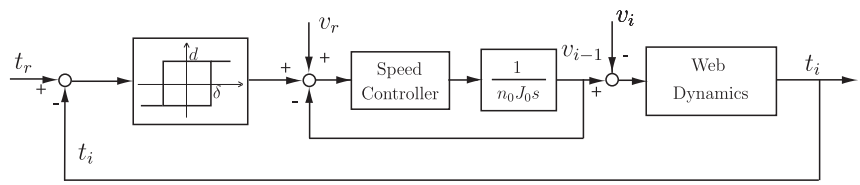

Fig. 8. Relay feedback in tension outer loop.

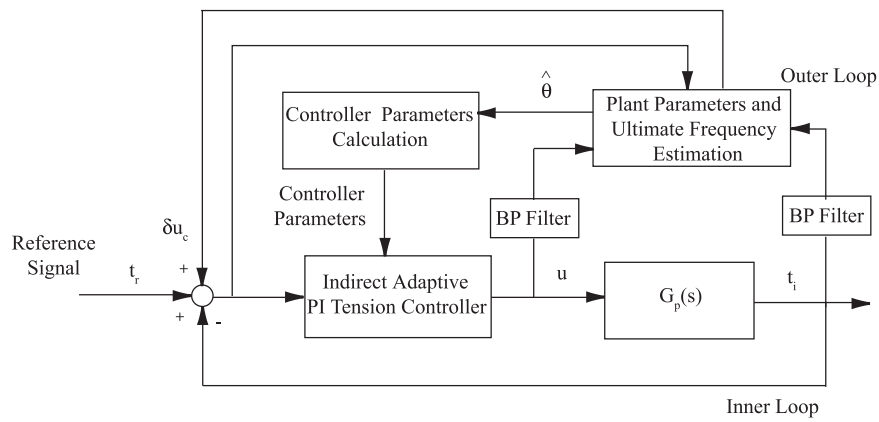

Fig. 9. Indirect adaptive PI tension control strategy.

signals is the phase ϕ. Based on this estimate of the plant transfer function the indirect adaptive PI controller gains are given by

$$k_p = \frac{\beta_1}{\alpha} \tag{21a}$$

$$T_i = \frac{\beta_2}{\omega_A} \tag{21b}$$

where β_1 and β_2 are tunable gains. The indirect adaptive PI control law is given by

$$u = k_p\left(1 + \frac{1}{T_i s}\right) \tag{22}$$

The ultimate frequency is obtained by the relay feedback technique a priori and is fixed for real-time estimation of the plant transfer function. If there are changes in the plant such as changes in the material roll diameter, then the ultimate frequency found a priori is no longer relevant. In the following we describe a method to automatically estimate the ultimate frequency online using an algorithm provided in [3]. The method is illustrated in Fig. 10. The ultimate frequency is estimated by injecting a small sinusoidal perturbation to the tension reference. The adaptation law for the ultimate frequency is

$$\dot{\omega} = \gamma \cos(\omega T)(u_0 - t_i) \tag{23}$$

where γ is the estimation gain and $u_0 = t_r + \delta u_c$ is the perturbed reference.

The phase shift of the plant is calculated based on the estimated parameters as

$$\phi = \tan^{-1}\left(\frac{\hat{b}_1 \sin(\hat{\omega}T)}{\hat{b}_1 \cos(\hat{\omega}T) + \hat{b}_2}\right) - 2\hat{\omega}T \tag{24}$$

where $\hat{b}_1$ and $\hat{b}_2$ are estimated plant parameters and $\hat{\omega}$ is the estimated ultimate frequency. The amplitude of the plant is calculated based on the estimated plant parameters and the phase shift (24) as

$$\hat{\alpha} = \frac{\hat{b}_1 \sin(\hat{\omega}T)}{\sin(2\hat{\omega}T + \phi)}. \tag{25}$$

The controller gains given in (21) now use the estimates $\hat{\omega}$ and $\hat{\alpha}$.

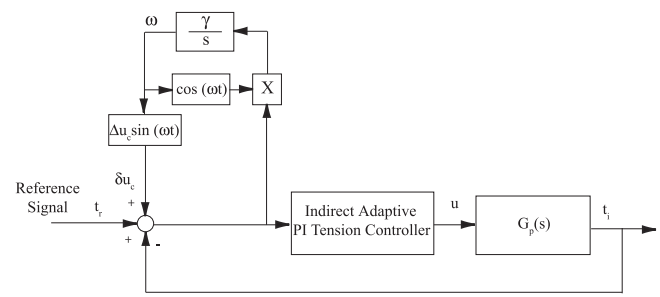

Fig. 10. Online estimation of ultimate frequency.

6. Experimental platform and procedure

A schematic of the R2R experimental setup used for experimentation is shown in Fig. 1. It is divided into four sections: unwind section, S-wrap section, pull-roll section, and rewind section. The five driven rollers in the machine are powered by AC motors, 15 HP (11.19 kW) for unwind and rewind and 5 HP (3.76 kW) for intermediate driven rollers. The S-wrap section, acts as the master speed section, which sets the web speed in the machine. The two driven rollers in the S-wrap section are under pure speed control. The unwind, pull-roll and rewind are under a speed-based tension control scheme as shown in Fig. 5. The cascaded control approach is common in industrial practice where the outer tension loop provides correction to the inner loop velocity reference. The outer tension loop is implemented in an external controller (ControlLogix controller from Allen Bradley) and the inner speed loop is implemented in the motor drive (Powerflex 755 AC drive from Allen Bradley). Encoders on motor shafts provide angular position feedback directly to the motor drives. The tension control algorithm is programmed in RSLogix 5000 software (Rockwell Automation PLC software). The real-time architecture used to run the R2R machine also includes analog/digital input/output modules and network boards. All the real-time hardware components of the machine are connected through a ControlNet communication network. The network is updated every 5 ms (Network Update Time) and data is communicated to the network every 10 ms (Request Package Interval). The lateral

guides, which control the web lateral position, are controlled by dedicated controllers independent of the ControlLogix real-time architecture.

The two adaptive PI control schemes presented in the previous two sections are implemented for control of tension in the unwind section of the R2R experimental platform. The tension feedback is obtained using load cells mounted on roller R8 shown in Fig. 1. The reference web tension is 20 lbf (89 N). In the first set of experiments, tension regulation experiments are carried out at the line speed of 100 FPM (0.51 m/s) for two web materials a Polyester film and Tyvek (a polymer material manufactured by Dupont). In the second set of experiments, the web line is accelerated to 150 FPM (0.76 m/s) and 250 FPM (1.02 m/s), then decelerated to 100 FPM (0.51 m/s), to evaluate the performance of the designed algorithms for changing speeds. The nominal values of the key web material parameters are given in Table 1.

For the MRA-PI control scheme, the reference model for the web tension system is selected based on model simulations:

$$G_{mt}(s) = \frac{15(1/1.2s+1)(1/50s+1)}{(1/12s+1)(1/5s+1)(s+5)(s+3)}. \tag{26}$$

The implementation of MRA-PI controller scheme is shown in Fig. 11. In Fig. 11, $F_p(s)$ and $F_i(s)$ are filters given by Eqs. (13) and (14), respectively.

For the indirect adaptive PI scheme based on automatic initialization, the initial plant parameter values are chosen as $b_1 = 0.5$ and $b_2 = 0.5$. The proportional and integral gain values are initialized with ultimate frequency obtained by relay feedback experiment for Tyvek and Polyester web materials. The adjustable parameters of the controller are selected empirically based on the performance of the system as $\beta_1 = 0.5$ and $\beta_2 = 0.2$. In the following we provide step-by-step guidelines for the implementation of the two adaptive schemes.

Table 1
Web material parameters.

Parameter	Numerical values	
	Tyvek	Polyester
Modulus (E)	9.3×10^4 psi (641.2 N/mm^2)	4.6×10^5 psi (3171.6 N/mm^2)
Width (w)	6 in (152.4 mm)	4.25 in (107.95 mm)
Thickness (h_w)	0.005 in (0.127 mm)	0.0038 in (0.097 mm)

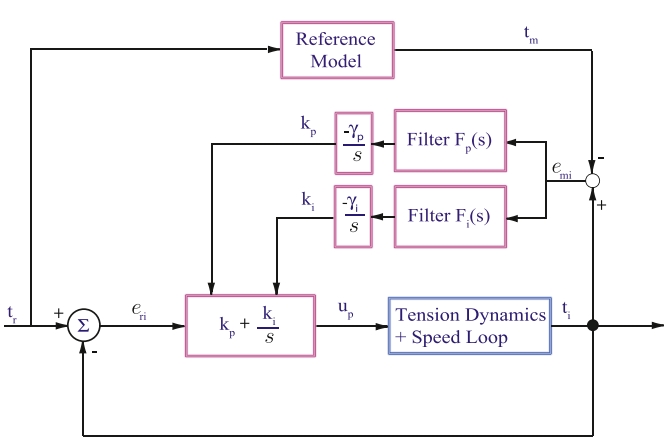

Fig. 11. Implementation of MRA-PI control.

6.1. Implementation guidelines for MRA-PI controller

Step 1: Select reference model parameters given in Eq. (8) based on the desired tension loop performance.
Step 2: Choose initial controller parameter estimates that satisfy the stability constraint criteria. The lower limit on the controller parameters can be imposed based on the constraint criteria.
Step 3: Estimate the controller gains online and generate controller output.
Step 4: The initial value of the proportional and integral gains may be increased in steps until the desired response is achieved.
Step 5: If the estimated controller parameters drift, an upper limit can be imposed on the estimates with suitable value that can be evaluated empirically or a method to project the parameters into a bounded set can be employed.

6.2. Implementation guidelines for indirect adaptive PI controller

Step 1: Evaluate the plant ultimate frequency offline by injecting relay oscillations in the system under tension but at zero speed.
Step 2: Initialize the integral gain with the ultimate frequency evaluated offline.
Step 3: Initialize the second order plant parameters. Then initialize proportional gain based on the ultimate frequency and plant parameters initial values.
Step 4: Filter the plant output and control input using the band pass filter at the estimated ultimate frequency.
Step 5: Estimate the ultimate frequency and plant parameters online for to account for dynamic changes in the plant.
Step 6: Evaluate the phase shift and amplitude ratio of the plant based on the estimated plant parameters and ultimate frequency.
Step 7: Select the adjustable controller parameters β_1 and β_2, and update the proportional and integral gains based on the estimated amplitude and ultimate frequency.

7. Experimental results

The tension response with a well-tuned fixed gain PI controller at 100 FPM (0.51 m/s) for Tyvek and Polyester web materials are shown in Fig. 12. The standard deviation is calculated for tension data in the steady-state and used as a performance metric. The standard deviation values indicate that at steady-state the web tension varies between ± 0.51 lbf for Tyvek web and ± 0.71 lbf for the Polyester web. The fixed speed PI controller gains are tuned based on a desired second-order characteristic equation that has the damping ratio $\xi = 1.1$ and the natural frequency $\omega = 4$ Hz as the design parameters for the selection of the PI gains. The outer tension loop fixed PI gains are tuned based on the second-order closed-loop speed dynamics and the first-order tension transfer function. The tuning was further fine-tuned empirically during experimentation. The following are tuned values for the fixed gains of the PI controller: the proportional and integral gain values are 25 and 0.5 for Tyvek and 39 and 0.75 for Polyester material. Since the elastic modulus of Polyester material is larger than Tyvek material, larger proportional and integral gains are needed for the Polyester material for a similar performance as that of Tyvek.

The tension response with the MRA PI control scheme at 100 FPM (0.51 m/s) for Tyvek and Polyester web materials are shown in Fig. 13. The same initial values of the controller parameter estimates and design parameters are used for both web materials.

The standard deviation values for the web tension in the unwind section at steady state vary between ± 0.53 lbf for Tyvek web and ± 0.66 lbf for Polyester web; these values are comparable or better than what is observed with a well-tuned fixed gain PI controller. Further, the MRA-PI scheme does not require tuning if different web materials are transported in the same R2R process line. This provides significant benefits to many R2R industries as tedious tuning is not required when processing different materials. The evolution of the estimates of controller parameters are shown in Figs. 14 through 15. Fig. 16 shows the performance of the MRA-PI algorithm to acceleration and deceleration of the web line at different speeds (AB: 100 FPM, BC: 150 FPM, CD: 250 FPM, and DE: 100 FPM) for the Tyvek web. The controller gain estimates adapt to compensate for the speed changes.

In the implementation of the indirect adaptive PI controller, first the relay feedback is introduced at zero web speed and when the tension is at its reference value. The relay amplitude is selected as 0.9, which is about 5% of the reference web tension of 20 lbf (89 N); this selection follows the guidance given on the selection of relay amplitude in the literature. The hysteresis is selected to be 0.006, based on the observation of the tension error fluctuation range. The relay feedback is given for short period of time to obtain sustained oscillations. The tension response data with oscillations is shown in Fig. 17. A close-up view of the relay output and the tension response for the Tyvek web is shown in Fig. 18. The amplitude of tension oscillations measured is 0.9. The period of tension oscillations is 2.6 s. So, the frequency of tension oscillations (ultimate frequency) for Tyvek web is 2.42 rad/s.

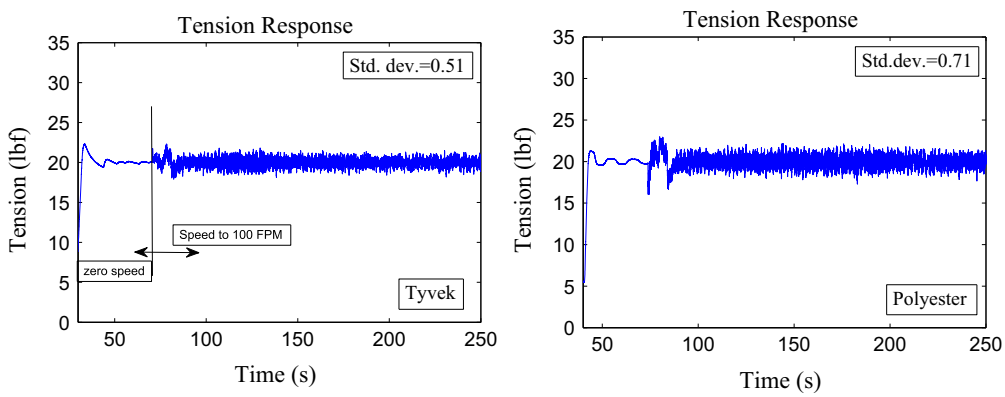

Fig. 12. Tension response at 100 FPM with a well-tuned industrial PI controller; Left: Tyvek; Right: Polyester.

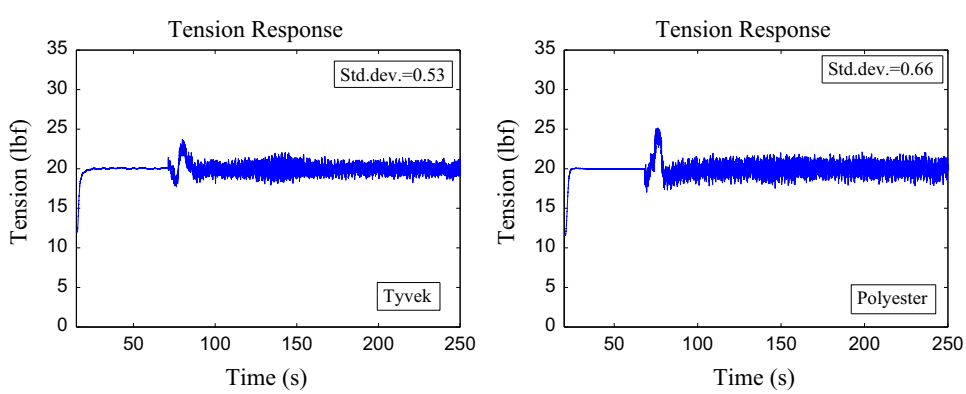

Fig. 13. Tension response at 100 FPM with MRA-PI controller; Left: Tyvek; Right: Polyester.

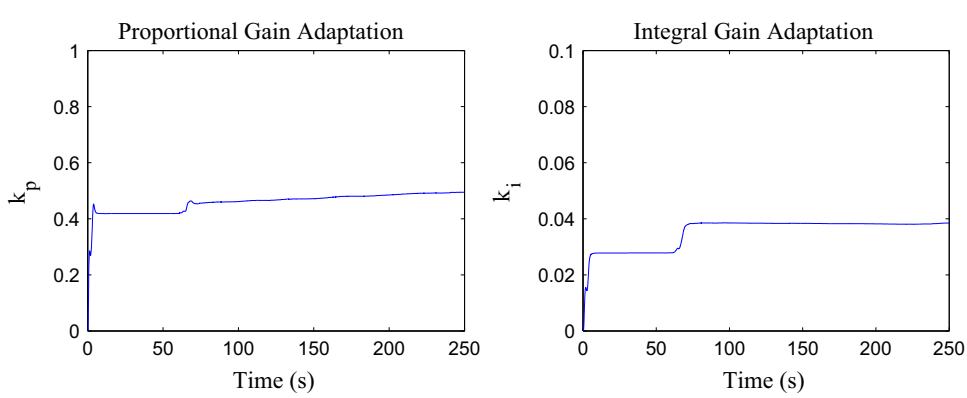

Fig. 14. Proportional and integral gain adaptation with MRA-PI controller for Tyvek web.

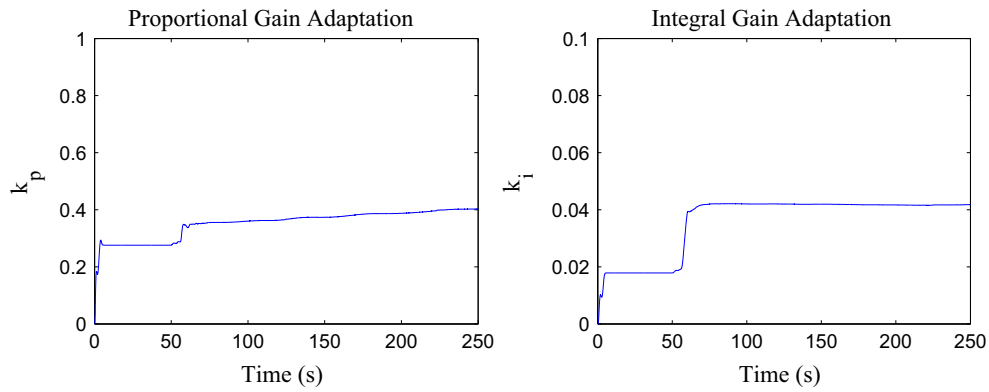

Fig. 15. Proportional and integral gain adaptation with MRA-PI controller for polyester web.

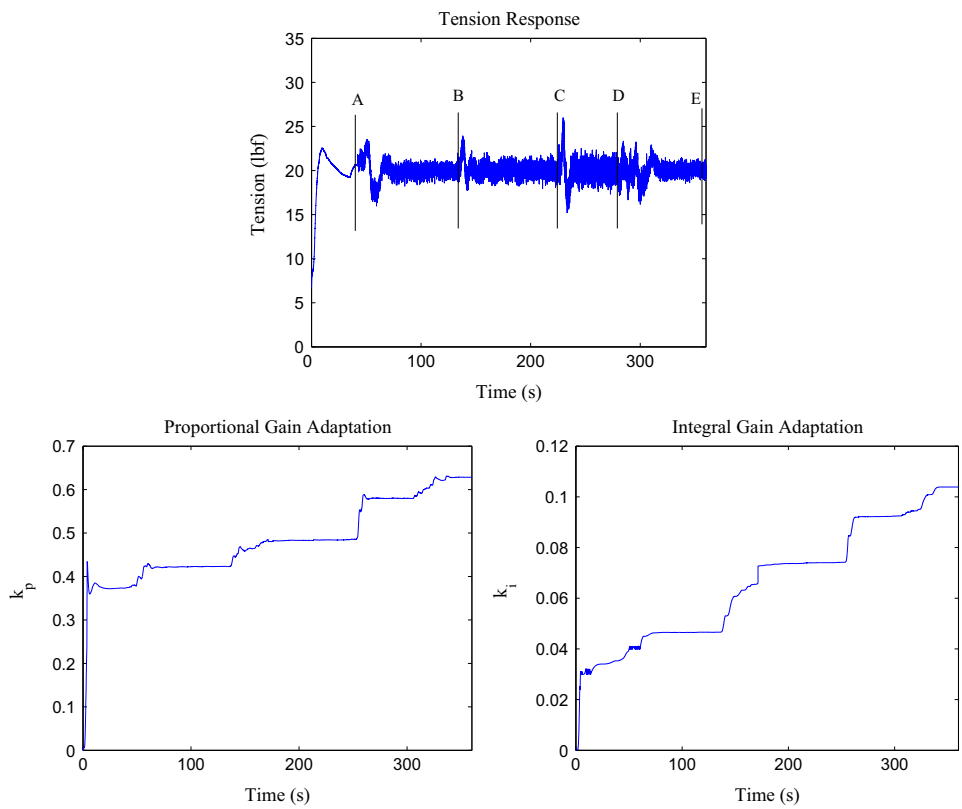

Fig. 16. Tension response and gain adaptation with MRA-PI controller for line speed changes.

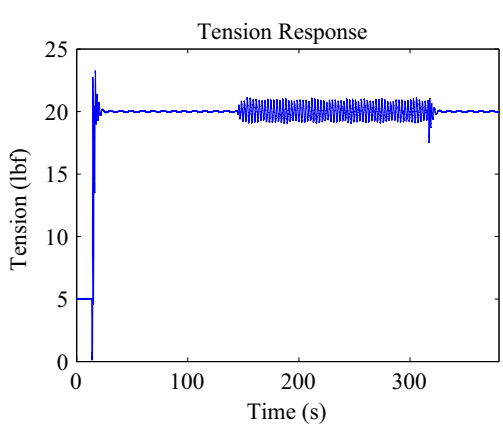

Fig. 17. Tension response to relay feedback for Tyvek web.

A similar experiment is performed for the Polyester web, which gave the frequency of tension oscillations (ultimate frequency) to be 2.07 rad/s. Adaptive gains are initialized with the ultimate frequency obtained by the relay feedback technique as formulated in Eqs. (24), (25) and (21). Once the controller estimates are initialized, non-zero speed experiments are performed to regulate tension.

The tension response with the indirect adaptive PI control scheme at 100 FPM (0.51 m/s) for Tyvek and Polyester web materials are shown in Fig. 19. The standard deviation values are ± 0.54 lbf for Tyvek web and ± 0.75 lbf for Polyester web. The evolution of the estimates of the controller parameters are shown in Figs. 20 through 21. Fig. 22 shows the performance of indirect adaptive PI algorithm to the acceleration and deceleration of the line (AB: 100 FPM, BC: 150 FPM, CD: 250 FPM, and DE: 100 FPM) for Tyvek web. The controller gains adapt to compensate for the changing speed.

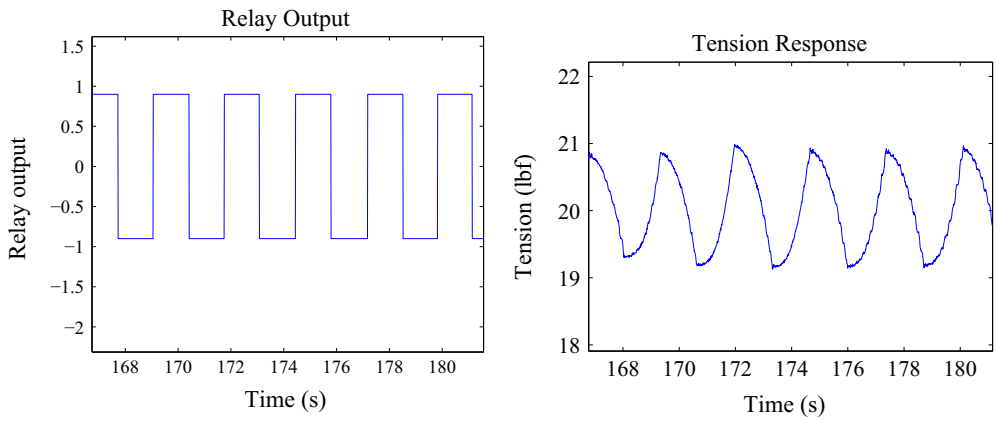

Fig. 18. Relay output and oscillating tension response for Tyvek web.

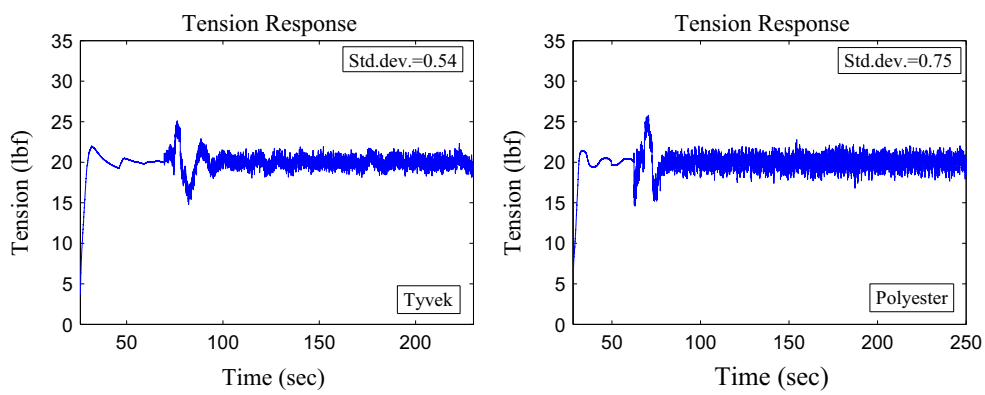

Fig. 19. Tension response at 100 FPM with indirect adaptive PI controller.

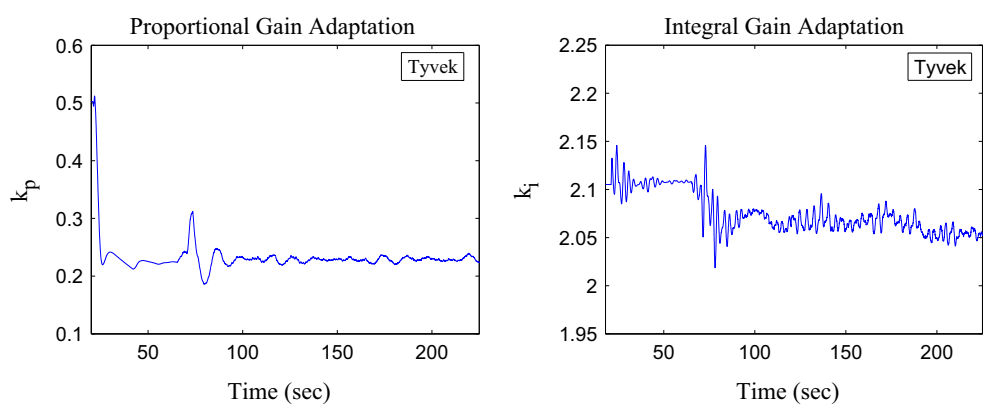

Fig. 20. Proportional and integral gain adaptation with indirect adaptive PI controller for Tyvek web.

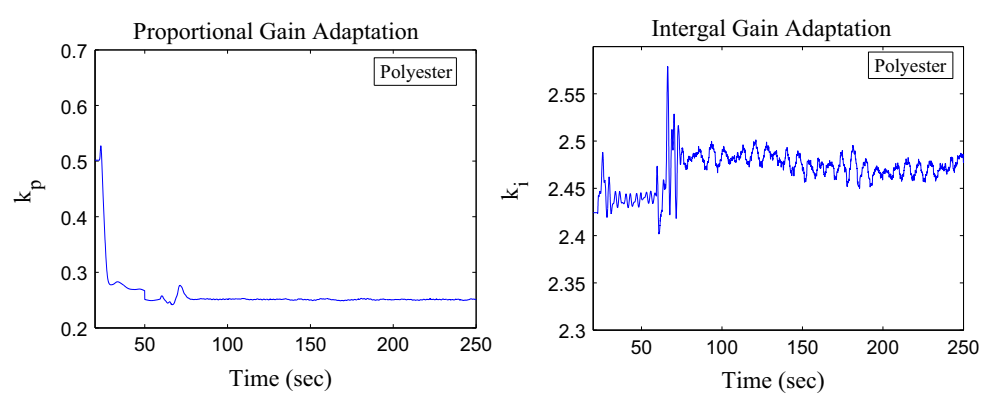

Fig. 21. Proportional and integral gain adaptation with indirect adaptive PI controller for polyester web.

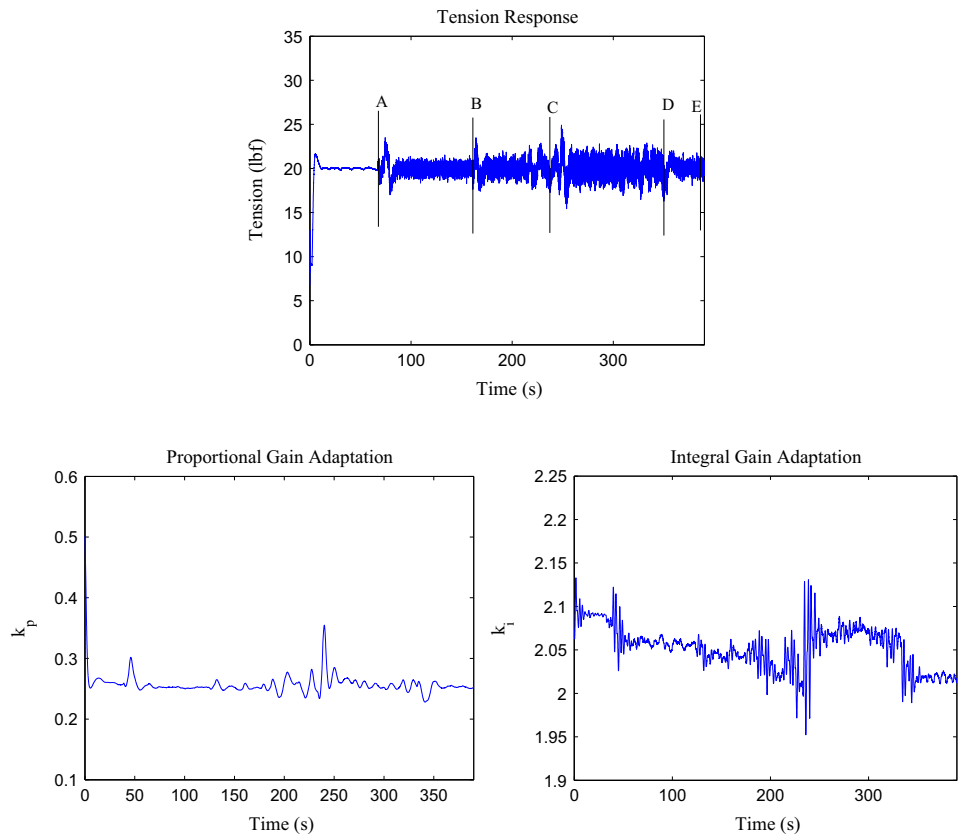

Fig. 22. Tension response and gain adaptation with indirect adaptive PI controller for line speed changes.

Both the MRA-PI and indirect adaptive PI controllers provide similar or better performance when compared to a well-tuned fixed gain PI controller. This performance is achieved without the need for any changes in controller gain initial estimates or design parameters when transporting different materials and during speed changes. A fixed gain PI requires tuning of controller gains for each material as well as for different speeds. Among the two adaptive PI schemes, the implementation of the indirect adaptive PI scheme has several advantages over the MRA-PI scheme: (1) there is no need for nominal plant parameters; (2) reference model is not required; (3) the controller parameter estimates in indirect adaptive PI controller are automatically initialized based on ultimate frequency which can be estimated online. For the indirect adaptive PI scheme implementation, one has to implement the relay feedback to obtain the initial estimate of the ultimate frequency. This does not pose a problem in R2R systems because in practice the tension is generated in the web at zero speed (typically called as tension-on) and then line speed is increased to the desired along a given acceleration profile. The relay feedback technique to obtain initial value of the ultimate frequency can be implemented in a straightforward fashion during the tension-on phase of the system operation. Further, all steps of the indirect adaptive PI scheme can be automated without the need for intervention.

8. Conclusions and future work

Design and implementation of two adaptive PI control schemes that automate the tension control systems in R2R manufacturing are discussed in this paper. The direct adaptive PI scheme based on the reference model requires some knowledge of the nominal parameters for initialization of the adaptive PI gains, whereas the adaptive gains in the indirect adaptive PI scheme can be automatically initialized using the relay feedback technique and an real-time algorithm to update the ultimate frequency. Both adaptive schemes work well for different operating conditions and materials without the need for extensive tuning. The main contribution of the paper is in developing simple adaptive PI schemes that can significantly automate the existing operation by replacing the fixed gain PI tension control schemes. The provided adaptive algorithms are of substantial benefit to the emerging R2R industry of printing electronics on moving flexible substrates where precise longitudinal and lateral positioning of the web is critical in addition to tension control.

There are further improvements that can be made to the adaptive PI schemes by considering the changes to web material properties during transport for various processes employed in R2R manufacturing. For example, there is need to develop suitable tension control systems in heating/cooling sections of the web line where the material properties may change significantly and it may not be possible to measure tension accurately; heating or cooling a web is common in many R2R processes such as printing, coating, lamination, and annealing. There is also a need to study the performance of these adaptive PI tension control schemes when dancer systems are used to measure tension.

Acknowledgements

This work was supported by the Web Handling Research Center at Oklahoma State University and the National Science Foundation under grant numbers 0854612 and 1246854.

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